Designing algorithms: flowcharts and pseudocode

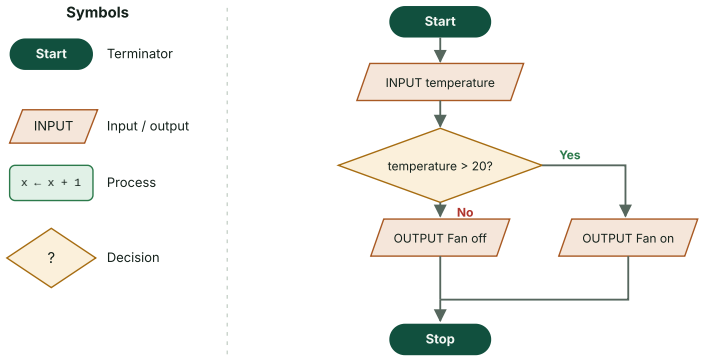
D10 FACT FILE

Specification: students should be able to

- Design solutions using algorithms, flow diagrams and pseudo-code
- Refine a solution to a problem during design

CCEA fact file: Digital Design Principles

Key words: algorithm, flowchart, terminator, process, decision, input/output, pseudocode, sequence, selection, iteration, refinement



Flowchart symbols and a flowchart that turns a fan on above 20 degrees

Representing algorithms

An algorithm is a sequence of precise steps that solves a problem. Flowcharts show it graphically using standard symbols: terminators for start/stop, parallelograms for input/output, rectangles for processes, diamonds for decisions and arrows for flow. Pseudocode describes it in structured English using keywords such as INPUT, OUTPUT, IF, WHILE and FOR, without worrying about a particular language's syntax.

Refinement

Designs are improved iteratively. Dry-running with test values exposes missing steps and logic errors before any code is written; feedback from the user can also add requirements. Refinements might add validation, replace repeated statements with a loop, or split a long algorithm into subroutines.

T Key terms from the mark schemes

Algorithm
A step-by-step plan for solving a problem.
Credited: U4 2019 Q6, 2025 Q3

Flowchart
Diagram of an algorithm using standard symbols for start/stop, input/output, process and decision.
Credited: U4 2022 Q8, 2023 Q4

Pseudocode
Language-independent, English-like instructions describing an algorithm.
Credited: U4 2024 Q7

Exam focus

Writing or completing algorithms is the largest single source of marks on Unit 4 (often 30+ marks). Follow every bullet point in the question, use the variable names given, and make the start and end of each IF and loop clear (2025 examiners). Flowchart questions: 2022 Q8(a), 2023 Q4(b). Defining pseudocode: 2024 Q7(b).

Unit 4 2024 Q7 [35]	Arrays, pseudocode, FOR/WHILE totals, minimum, bubble sort, linear and binary search
Unit 4 2023 Q4 [23]	Decomposition and abstraction; flowchart symbols; syntax/logic errors; functions; menu validation loop
Unit 4 2022 Q8 [9]	Flowchart selection and iteration; dry runs
Unit 4 2019 Q9 [19]	Test strategies, data types, validation loop, design, discount selection

? Check yourself

1. Define the term algorithm. [2]
2. Describe what is meant by pseudocode. [2]
3. Name and describe the purpose of two flowchart symbols. [4]

Answers: 1. A precise, step-by-step set of instructions (1) / to solve a problem / carry out a task (1) 2. An English-like (structured English) description of an algorithm using programming-style keywords (1) / that is language-independent and ignores exact syntax, so it can be converted into any language (1) 3. Two symbols, each named (1) + purpose (1), e.g.: terminator (rounded rectangle) (1) – start/stop of the algorithm (1) / input/output (parallelogram) (1) – input a value or output a result (1) / process (rectangle) (1) – a calculation or assignment (1) / decision (diamond) (1) – a Yes/No question with two exits (1) / flow line (arrow) (1) – shows the direction of flow (1) / subroutine (double-sided rectangle) (1) – calls a predefined procedure/function (1)